

WIDEBAND MIMO ANTENNA WITH METASURFACE

Phase I: CPW-fed Wideband Antenna with a Split-Ring Metasurface Reflector



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OCTOBER 2026

Abstract

Fast wireless links need wideband antennas, and multiple-input multiple-output (MIMO) systems need several of them. A printed monopole gives a wide band but low gain, because it radiates to both sides of the board. A metal plate behind it raises the gain only when placed far away, which makes the antenna thick. This project places a thin metasurface close behind the antenna instead. This report covers Phase I. A coplanar waveguide (CPW) fed decagonal monopole was designed on a $50 \times 50 \times 1.6$ mm FR-4 board in CST Studio Suite 2019. Moving the ground closer to the patch and using a 15 mm patch radius gave S_{11} below -10 dB from 2.16 to 15.73 GHz. A 6×5 split-ring metasurface was placed 2 mm behind the antenna. The match holds over most of the band, but three narrow mismatch gaps appear between 3.0 and 5.6 GHz. With the metasurface, the peak directivity rises from 3.7 to 8.8 dBi, and the directivity is higher over about 78 % of 2–6 GHz. Realized gain over the whole band, metasurface tuning and a MIMO version are the next steps.

Introduction

Background and Motivation

Modern wireless systems need antennas that work over a wide range of frequencies. MIMO systems use several antennas at each end of the link to carry more data without more transmit power [1], [2]. A compact wideband antenna that can later be repeated in a MIMO layout is therefore useful [3].

Printed monopole antennas are a common choice for wideband use. A flat disc or polygon patch has several close resonances that join into one wide band [4], [5]. A CPW feed puts the feed line and the ground on the same side of the board, so no vias are needed [6]. The weak point of such an antenna is its gain. It radiates almost equally to the front and to the back, so much of the power is lost behind it [7].

A metal plate behind the antenna can turn the backward wave forward. However, a plain metal plate helps only when it is far from the antenna, so the antenna becomes thick. A metasurface is a thin board covered with many small, repeated metal cells that shape the reflected wave [8]. Because of this, it can sit much closer to the antenna. The split ring is a widely used metasurface cell [9]. Fig. 1 shows the structure proposed in this project.

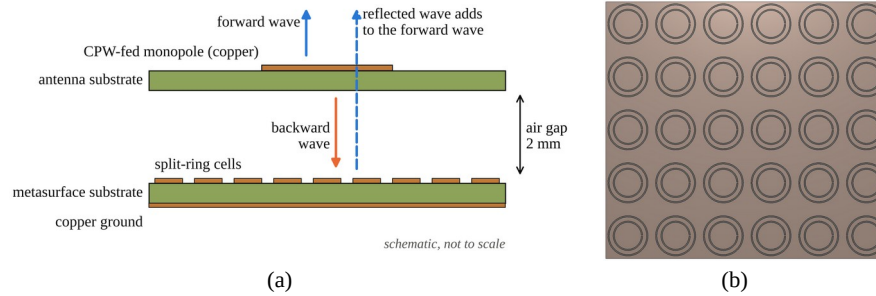


Fig. 1. Proposed structure: (a) side view of the antenna, the 2 mm air gap and the metasurface (schematic, not to scale), and (b) CST model of the 6×5 double split-ring metasurface, top view.

Problem Statement and Objectives

A printed monopole gives a wide band but low gain, and a metal reflector would make it thick. This project aims to build a thin wideband antenna with higher gain, and later a MIMO version of it. The objectives are:

- To design a CPW-fed wideband monopole antenna that covers about 2.2–15 GHz.
- To optimise the ground position and the patch size for the best impedance match.
- To add a split-ring metasurface 2 mm behind the antenna to raise its gain.
- To extend the design to a MIMO antenna, then fabricate and measure it (Phase II).

Literature Review

Several groups have placed a reflector behind a wideband printed antenna to raise its gain. Table 1 lists the works closest to this project. Sen et al. [10] placed a metasurface of double split rings behind a circular monopole. The split angle changes from column to column, and the gain rose by about 5.5 dB. Al-Gburi et al. [11] placed a CPW-fed ring monopole over a 19×19 loop frequency selective surface (FSS) with a ground plane. The peak gain rose from 6.7 to 11.5 dBi, and the whole structure is 10 mm thick. Hussain et al. [12] put a 5×5 FSS 9 mm behind a CPW-fed hexagonal patch, and the peak gain rose from 6.5 to 10.5 dBi. Hammache et al. [13] used a 7×7 FSS 20 mm behind a CPW-fed hexagonal monopole, and the gain rose from 2.2 to 8.4 dBi.

Metasurfaces are also used in MIMO antennas. Hasan et al. [14] placed a copper-backed 10×10 split-ring metasurface 12 mm behind a 4-port antenna for 3.08–7.75 GHz. The gain rose from 5.4 to 8.3 dBi, and the isolation between ports stayed above 15.5 dB. Wu et al. [15] used a polarization-conversion metasurface in a 2-port antenna for 4.76–6.77 GHz, with a gain of 7.95 dBic and isolation above 19.8 dB.

These works show that a reflector or metasurface can raise the gain of a printed antenna by several dB. In most of them, however, the reflector sits far behind the antenna. From them, this project takes three ideas: a CPW-fed monopole as the radiator [11]–[13], split-ring cells for the metasurface [10], [14], and a copper-backed metasurface behind a MIMO antenna for Phase II [14].

Table 1. Printed antennas with a reflector or metasurface behind them, compared with this work. Gain is the peak gain without → with the reflector.

Reference	Antenna	Reflector	Gap	Peak gain
Sen et al. (2017) [10]	Circular monopole	Double split-ring metasurface (graded splits)	Not given in abstract	About +5.5 dB
Al-Gburi et al. (2022) [11]	CPW-fed ring monopole	19×19 loop FSS with ground plane	10 mm (total height)	6.7 → 11.5 dBi
Hussain et al. (2023) [12]	CPW-fed hexagonal patch	5×5 ring-frame FSS	9 mm	6.5 → 10.5 dBi
Hammache et al. (2024) [13]	CPW-fed hexagonal monopole	7×7 FSS	20 mm	2.2 → 8.4 dBi
Hasan et al. (2022) [14]	4-port MIMO patch (3.08–7.75 GHz)	10×10 copper-backed split-ring metasurface	12 mm	5.4 → 8.3 dBi; isolation > 15.5 dB
This work	CPW-fed decagonal monopole	6×5 double split-ring metasurface, copper-backed	2 mm	3.7 → 8.8 dBi (directivity, 1–6 GHz)

Research Gap

In the reviewed works, the reflector usually sits 9–20 mm behind the antenna [11]–[14]. This makes the antenna thick. A much thinner metasurface, only a few millimetres behind the antenna, working over a band as wide as about 2–15 GHz, is rarely reported. The reviewed metasurface MIMO antennas also cover narrower bands [14], [15]. This project therefore aims to place a split-ring metasurface only 2 mm behind a wideband antenna, and then to build a MIMO version of it.

Methodology / Proposed Work

1. Simulation Setup

All simulations use CST Studio Suite 2019 (frequency-domain solver, 0–18 GHz). The antenna is printed in copper on a $50 \times 50 \times 1.6$ mm FR-4 board. The radiator is a decagonal (10-sided) patch of circumradius R , with its centre fixed at 8 mm on the feed axis. The CPW feed has a 3.0 mm signal strip and 0.5 mm slots, and a waveguide port referenced to 50Ω excites it. L_g is the position of the ground edge: -20 mm puts the ground far from the patch, and -7 mm brings it close.

2. Design Workflow

Fig. 2 shows the design steps. Steps 1–7 are done, and step 8 is Phase II.

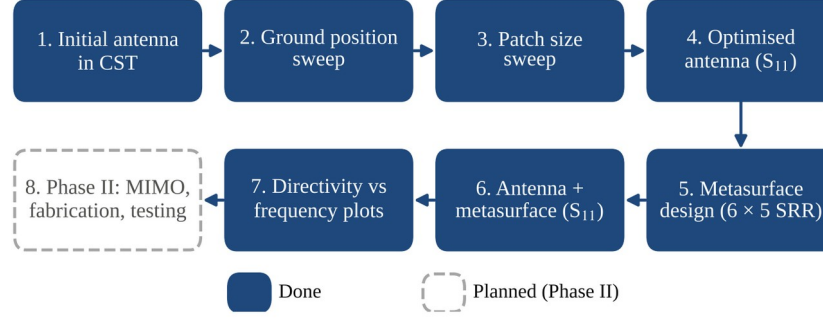


Fig. 2. Design workflow (dark: done; dashed: Phase II).

3. Initial Antenna

The first design used $R = 15$ mm and $L_g = -20$ mm (Fig. 3(a)), which leaves a 13.73 mm gap between the ground and the patch. Its match is poor (Fig. 3(b)): from 2 to 7.7 GHz, S_{11} stays between about -2.3 and -7.5 dB. Above 7.7 GHz there are only a few narrow dips below -10 dB, the deepest about -45 dB at 8.56 GHz.

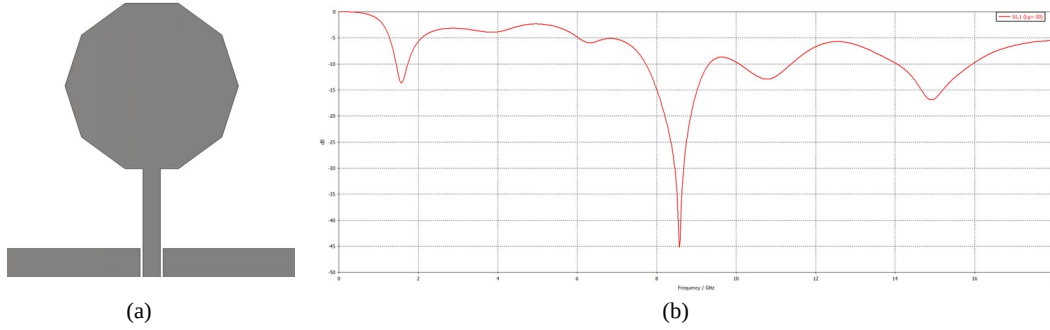


Fig. 3. Initial antenna ($R = 15$ mm, $L_g = -20$ mm): (a) CST model and (b) simulated S_{11} .

4. Ground Position and Patch Size Sweeps

First, L_g was swept from -20 to -7 mm in 10 values, with $R = 15$ mm (Fig. 4(a)). Moving the ground closer to the patch strengthens their coupling and improves the low-frequency match. Only $L_g = -7$ mm keeps S_{11} below -10 dB across the band, so it was chosen; the feed gap is then about 0.73 mm.

Next, R was swept from 4 to 15 mm in 13 values, with $L_g = -7$ mm (Fig. 4(b)). Every patch size gives a first dip near 2.3–2.7 GHz, but small patches leave much of 3–12 GHz poorly matched (-3 to -9 dB). Only $R = 15$ mm joins the resonances into one continuous band. Because the patch centre is fixed, a larger R also narrows the feed gap.

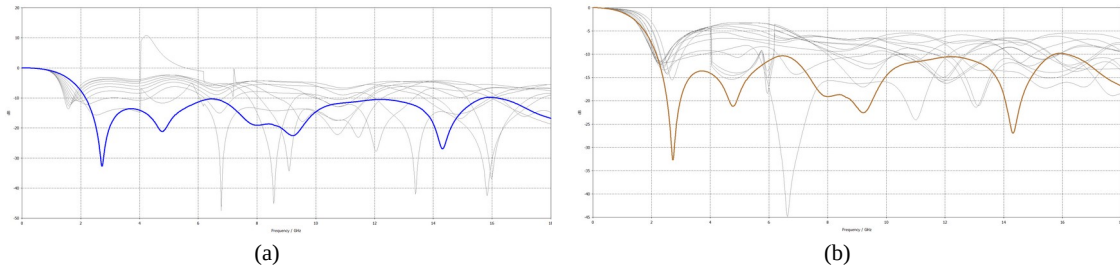


Fig. 4. S_{11} sweeps: (a) L_g from -20 to -7 mm (blue: -7 mm) and (b) R from 4 to 15 mm (brown: 15 mm); other values are grey. Jumps near 4.0 and 6.2 GHz in some grey curves are sampling artefacts.

5. Optimised Antenna

The optimised antenna ($R = 15$ mm, $L_g = -7$ mm) is shown in Fig. 5(a). Its S_{11} is below -10 dB from 2.16 to 15.73 GHz, about 13.6 GHz of bandwidth (Fig. 5(b)). It has resonances at 2.73, 4.78, 9.23 and 14.32 GHz; the deepest is -32.65 dB at 2.73 GHz. The match is thin near 6.45 GHz (-10.3 dB) and 12.2 GHz (-10.5 dB).

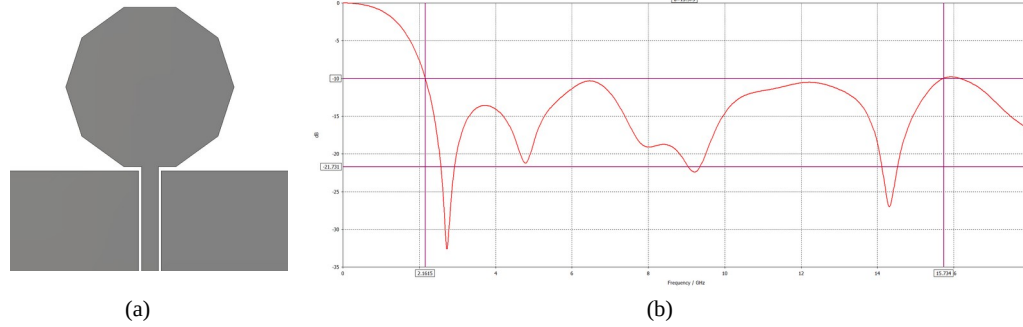


Fig. 5. Optimised antenna ($R = 15$ mm, $L_g = -7$ mm): (a) CST model, front view, and (b) simulated S_{11} with markers at the -10 dB band edges (2.16 and 15.73 GHz).

6. Metasurface Design

The metasurface is a 6×5 array of double split-ring cells, each with two concentric split rings, printed on FR-4 with full copper on the back (Fig. 1(b)). It is placed 2 mm behind the antenna across an air gap (Fig. 1(a)). The reflection phase of a single cell has not been simulated yet.

7. Antenna with Metasurface

With the metasurface, S_{11} is below -10 dB from about 2.0 GHz up to 18 GHz, the end of the simulation, except in three narrow gaps: 3.0–3.4 GHz (worst -6.6 dB near 3.14 GHz), 4.5–4.7 GHz (worst -9.7 dB) and 5.1–5.6 GHz (worst -8.8 dB near 5.34 GHz) (Fig. 6). At such a small gap, the metasurface loads the antenna and changes its input match. The gap, ring size and ground position will be tuned to remove these gaps. Subsection 8 shows how the metasurface changes the directivity.

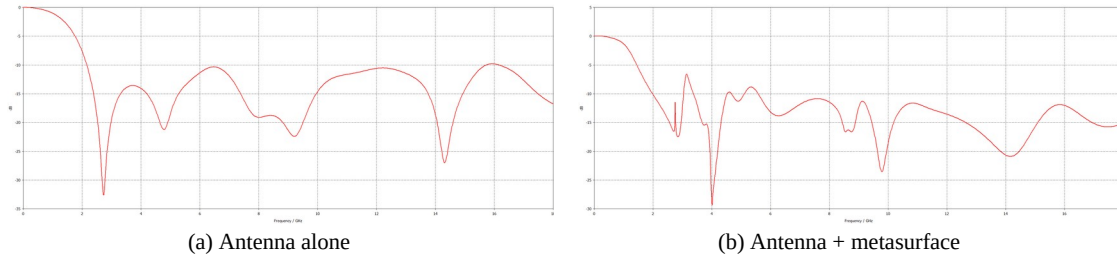


Fig. 6. Simulated S_{11} : (a) antenna alone and (b) antenna with the metasurface 2 mm behind it.

8. Directivity with and without the Metasurface

Fig. 7 compares the directivity of the antenna alone and with the metasurface, taken as the highest value in the $\phi = 0^\circ$ plane. Directivity is the gain without losses. The metasurface run covers only 1 to 6 GHz, so nothing is known above 6 GHz yet. With the metasurface, the peak directivity rises from 3.7 dBi near 4 GHz to 8.8 dBi at 5.9 GHz. Over 2–6 GHz (below 2 GHz the antenna is not matched), the directivity with the metasurface is higher on about 78 % of the band, and its average rises from 2.3 to 3.9 dBi (+1.6 dB).

The antenna alone is higher in only two narrow bands: about 2.9–3.1 GHz (by up to 3.1 dB, next to the first mismatch gap) and 3.3–4.0 GHz. Overall, the metasurface makes the antenna more directive. Its copper back reflects the backward wave forward, so more power goes forward. This is why it was added. Realized gain, which also counts losses and mismatch, is the next step.

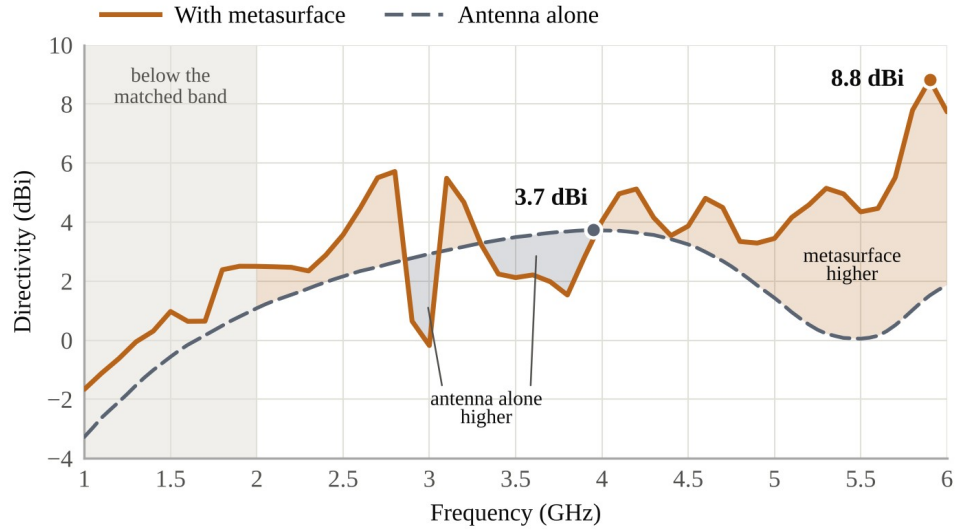


Fig. 7. Simulated directivity (highest value in the $\phi = 0^\circ$ plane) of the antenna alone and with the metasurface 2 mm behind it, redrawn from the CST plots. Shading marks where each case is higher.

Work Done Till Mid-Semester

In Phase I so far, the single antenna was designed and optimised in CST, the split-ring metasurface was designed and simulated with it, and the directivity with and without the metasurface was compared. Table 2 gives the status of each Phase I task and its main result.

Table 2. Status of Phase I.

Task	Status	Main result
Initial antenna ($R = 15$ mm, $L_g = -20$ mm)	Done	Poor match from 2 to 7.7 GHz
Ground position sweep (L_g from -20 to -7 mm)	Done	$L_g = -7$ mm chosen
Patch size sweep (R from 4 to 15 mm)	Done	$R = 15$ mm chosen
Optimised antenna, S_{11}	Done	Below -10 dB from 2.16 to 15.73 GHz
Metasurface design (6×5 double split rings)	Done	Placed 2 mm behind the antenna
Antenna + metasurface, S_{11}	Done	Below -10 dB from about 2.0 to 18 GHz, except 3.0–3.4, 4.5–4.7 and 5.1–5.6 GHz
Directivity vs frequency, antenna alone	Done	Peak 3.7 dBi near 4 GHz
Directivity vs frequency, antenna + metasurface	Done	Peak 8.8 dBi; higher over 78 % of 2–6 GHz
Realized gain vs frequency (whole band)	Remaining	—
Metasurface tuning to remove the mismatch gaps	Remaining	—

Challenges

- **Long run times:** the sweeps and the antenna + metasurface model are slow to simulate.
- **Sweep artefacts:** some sweep curves jump near 4.0 and 6.2 GHz, and one goes above 0 dB, which is not physical. They come from too few frequency samples, not from the mesh.
- **Thin match margin:** the optimised antenna is only just matched near 6.45 GHz (-10.3 dB) and 12.2 GHz (-10.5 dB), so small fabrication errors could push these points above -10 dB.
- **Metasurface detuning:** at a 2 mm gap the metasurface changes the input match (three narrow mismatch gaps between 3.0 and 5.6 GHz) and lowers the directivity in two narrow bands between 2.9 and 4.0 GHz.

Work Plan for Next Phase

The remaining Phase I work comes first:

- Plot realized gain (losses and mismatch included) vs frequency for both cases over the whole band, from 2 to 15 GHz, not only 1–6 GHz.

- Tune the metasurface (gap, ring size) and the ground position to remove the mismatch gaps and the directivity dips between 2.9 and 4.0 GHz, and re-run the sweeps with more samples.

Phase II will then cover the MIMO antenna and the hardware:

- Build a 4-port MIMO antenna from four copies, each rotated by 90°, backed by the metasurface.
- Check the isolation between ports and the envelope correlation coefficient (ECC).
- Fabricate the antenna and the metasurface on FR-4, and measure them with a vector network analyser (VNA) and in an anechoic chamber.

Expected Outcomes

- A simulated CPW-fed wideband antenna covering about 2.2–15 GHz, with realized gain vs frequency plots with and without the metasurface.
- A tuned split-ring metasurface 2 mm behind the antenna that keeps it matched across the band. The peak directivity already rises from 3.7 to 8.8 dBi; realized gain will confirm the increase.
- A 4-port MIMO version of the antenna with the metasurface, checked for isolation and ECC.
- A fabricated prototype whose measured S-parameters and radiation patterns are compared with simulation.

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